

Frequency-Informed Encoder-Decoder Models with Expressivity for Time Series Anomaly Detection

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Abstract—Improving anomaly detection accuracy in time series data is essential for the operation of communication networks. Encoder-decoder models, which learn the correlations within the data, have been widely used for anomaly detection because the models can be trained without anomalous data. To capture the correlations within the data, encoder-decoder models are trained to reconstruct the data by using their encoders and decoders. However, existing encoder-decoder models struggle to reconstruct the input data depending on the geometric features of the data. In this paper, we focus on periodic time series data with complex geometric features and propose an expressive encoder-decoder model with learnable parameters that can exactly reconstruct periodic time series data. In anomaly detection experiments on two periodic time series datasets, our proposed model achieved higher area under the curve (AUC) scores than existing encoder-decoder models.

Index Terms—Anomaly detection, Time series data, Encoder-decoder models, Exact reconstruction

I. INTRODUCTION

Anomaly detection of time series data is one of the most important tasks in the operation of communication networks. If operators miss anomalous network conditions, it could seriously impact our social activities [1]. Therefore, operators have to detect anomalous data accurately.

Encoder-decoder models, such as Autoencoder (AE) [2], [3], have been widely used for anomaly detection. In the training phase, encoder-decoder models are trained to reconstruct the training input data by first encoding the data into latent representations and then decoding the latent representations into the original data, which enables the models to learn the correlations within the data. In the test phase, the test data with large reconstruction errors are detected as anomalous. Since there are limited anomalous data and anomaly-labeled data in communication networks, encoder-decoder models, which can be trained without anomalous data, are suitable for anomaly detection.

However, existing encoder-decoder models have a limitation in reconstructing input data with complex geometric features, resulting in poor anomaly detection accuracy. One example of the geometric features is the *holes*, which we will explain in detail in Section II-B. The *holes* indicate that the correlations within the data are strongly non-linear, making encoder-decoder models fail to learn the correlations.

We focus on periodic time series data with the *holes* because real traffic data, such as traffic volume flowing through routers, often exhibit periodicity and non-linear correlations. One example of time series data that we focus on is periodic

time-shifted data, where time-shifts indicate that one time series is shifted relative to another time series. The time-shifts often occur in real traffic data due to differences in server locations and variations in the features of the traffic data. The shifted patterns imply the non-linear correlations within the time-shifted data, resulting in the presence of the *holes*.

In this paper, we propose an encoder-decoder model with expressivity, that is, learnable parameters that can exactly reconstruct periodic time series data even if the data have the *holes*, which we say the proposed model has the *ideal learnable parameters* for periodic time series data. We focus on the fact that periodic time series data can be reconstructed from their frequencies through the Fourier transform. Therefore, to enable the proposed encoder-decoder model to have the *ideal learnable parameters* for periodic time series data, we introduce the Affine with Frequency-Encoded (AFE) layer, which incorporates the frequencies and correlations within the data. Indeed, we theoretically show in Theorem III.1 that the encoder-decoder model using the AFE layer in its decoder has the *ideal learnable parameters* for periodic time series data.

To show the effectiveness of the proposed model experimentally, we conducted anomaly detection experiments on two periodic time series data. The first data are synthetic time series data with the *holes*. The second data are time-shifted data based on real-world traffic data. In both experiments, the proposed models outperformed existing models in terms of area under the curve (AUC) on anomaly detection. The experimental results demonstrate that the proposed model enhances anomaly detection accuracy in periodic time series data from communication network operations.

II. BACKGROUND

In this section, we first review encoder-decoder models that are commonly used in anomaly detection. Next, we discuss a limitation of encoder-decoder models for reconstruction, which stems from the geometric features of the data. Finally, we explain the related works that address the limitation and use frequencies in their encoder-decoder models.

A. Review of encoder-decoder models

An encoder-decoder model is composed of two elements: encoder and decoder. The encoder transforms the input data x into the latent representation z that captures the correlations within the data. The decoder transforms the latent representation

tation z into output data $\hat{x}$ so that $\hat{x}$ becomes close to the original input data x .

Encoder-decoder models vary based on the architectures of their encoders and decoders. The following models are often used in anomaly detection. Autoencoder (AE) [2] is the simplest encoder-decoder model. The encoder and decoder are constructed with compositions of affine transformations and activation functions, which we call the Fully Connected (FC) layer. One-dimensional Convolutional Neural Network autoencoder (1DCNN-AE) [4], [5], Long Short-Term Memory autoencoder (LSTM-AE) [6], [7], and Transformer autoencoder (Transformer-AE) [8], [9] leverage the Convolutional layer, the LSTM layer, and the Attention layer in their encoders and decoders, respectively.

The above encoder-decoder models are used in Section IV as the existing models.

B. Limitation of encoder-decoder models for reconstruction

A limitation of existing encoder-decoder models is that they are not guaranteed to have the *ideal learnable parameters* when the data have many *holes*, which are rigorously defined as persistent homology [10].

One example of data with the *holes* is time-shifted data when plotted in Euclidean space. For instance, the time series data $\{(\cos 2\pi t, \sin 2\pi t)\}_{t \in \mathbb{R}}$ are typical time-shifted data. When $\{(\cos 2\pi t, \sin 2\pi t)\}_{t \in \mathbb{R}}$ are plotted in Euclidean space, the data form the circle $S^1 = \{(x_1, x_2) \in \mathbb{R}^2 | x_1^2 + x_2^2 = 1\}$, which have the *holes*.

When data have the *holes*, the correlations within the data are strongly non-linear, making encoder-decoder models fail to obtain latent representations and reconstruct the data. More mathematically, consider the case where you train an AE that encodes the circle $x \in S^1$ to a one-dimensional representation $z \in \mathbb{R}$ and then decodes z to the output $\hat{x}$, which we call S^1 -AE. We show that there does not exist an S^1 -AE that has the *ideal learnable parameters*. To prove this statement, we assume that the S^1 -AE has the *ideal learnable parameters* and derive a contradiction as follows. Denote the encoder and the decoder in the S^1 -AE by Enc and Dec , respectively. Let $x, y \in S^1$, $x \neq y$, be input data. Since we assume the S^1 -AE has the *ideal learnable parameters*, $x = Dec(Enc(x))$ and $y = Dec(Enc(y))$ hold. Therefore, with $x \neq y$, we obtain $Dec(Enc(x)) \neq Dec(Enc(y))$, implying $Enc(x) \neq Enc(y)$. Moreover, Enc is a continuous function because Enc is a composition of continuous functions. Hence, when we write $x \in S^1$ as $x = e^{i\theta}$, where i denotes the imaginary unit, the encoder $Enc(e^{i\theta})$ must either monotonically increase or monotonically decrease as θ increases from 0. This phenomenon is illustrated in Figure 1, implying that $Enc(e^{0i}) \neq Enc(e^{2\pi i})$. This is a contradiction because $e^{0i} = e^{2\pi i}$, concluding that there does not exist an S^1 -AE that has the *ideal learnable parameters*. From this example, we can see that encoder-decoder models are not guaranteed to have the *ideal learnable parameters* for data with the *holes*.

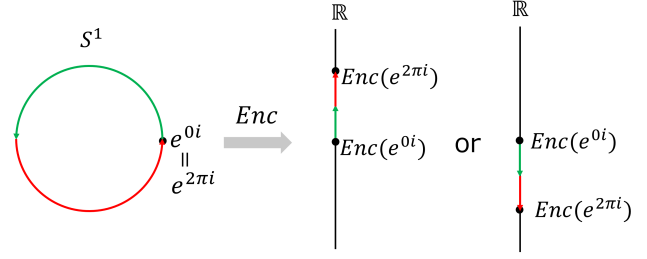


Fig. 1: The visualization of the encoder in the S^1 -AE with the *ideal learnable parameters*. Since $Enc(x) \neq Enc(y)$, the encoder must monotonically increase or monotonically decrease.

C. Related works

We review existing encoder-decoder models that leverage geometric features and models that use frequencies of time series data.

There are various works that focus on the geometric features of data, which are not necessarily time series data. Topological autoencoder [11] is the AE that incorporates the persistent homology of data into the loss function. Homeomorphic variational autoencoder [12] changes the latent space from Euclidean space to the manifold $SO(3)$. Liu et al. [13] proposed the graph attention network with topological features for time series anomaly detection.

In addition, several works used frequencies in their encoder-decoder models. Fedformer [14] is a transformer that uses frequencies. Yi et al. [15] constructed an encoder-decoder model in the frequency domain.

However, these works do not prove that their encoder-decoder models have the *ideal learnable parameters*.

III. PROPOSED MODEL

In this section, we propose an encoder-decoder model that has the *ideal learnable parameters* for periodic time series data. To ensure that the proposed model has the *ideal learnable parameters*, we introduce the Affine with Frequency-Encoded (AFE) layer that integrates both frequency information and correlations within the data.

In the following subsections, we first outline the architecture of the proposed model in Section III-A. Next, in Section III-B, we prove that the proposed model has the *ideal learnable parameters* for periodic time series data. Finally, in Section III-C, we discuss the characteristics of the proposed model.

A. Architecture of the proposed model

We show the overall outline of the proposed model in Figure 2. The input data of the proposed model are time series data x_t , which are encoded into their latent representations z_t . We can use any type of encoder such as FC layers, Convolutional layers, LSTM layers, and Attention layers. Then, the latent representations z_t are decoded into $\hat{x}_t$ to approximate the input x_t . The decoder is constructed by compositions of the AFE layers and activation functions.

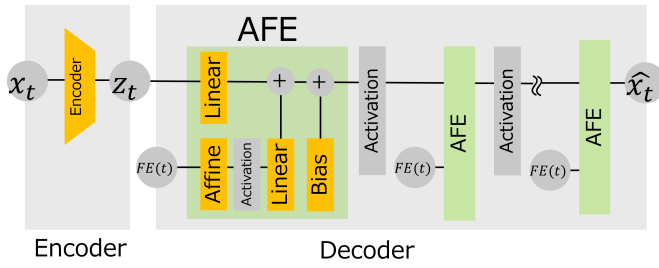


Fig. 2: The architecture of the proposed encoder-decoder model. The encoder is the same as that in usual encoder-decoder models. The decoder is constructed by compositions of the AFE layers and activation functions. The orange parts are learnable parameters.

The AFE layer is designed to incorporate the frequencies and correlations within time series data, as shown in Figure 2. To use the frequencies of time series data, we first define the Frequency-Encoding function $FE(t)$ by the following equation:

$$FE(t) = (\sin 2\pi f_1 t, \cos 2\pi f_1 t, \dots, \sin 2\pi f_M t, \cos 2\pi f_M t), \quad (1)$$

where $f_1, \dots, f_M$ are the top M values in magnitude from the power spectrum of the time series data, which are computed via the Fast Fourier Transform (FFT). Then, we construct the AFE layer by the following equation:

$$AFE(x, FE(t)) = W_1 x + W_2(\sigma(W_3 FE(t) + b_1)) + b_2, \quad (2)$$

where W_1, W_2 , and W_3 are weights, b_1 and b_2 are biases, and σ is an activation function. Based on the standard FC layer, the AFE layer transforms $FE(t)$ through a composition of an affine transformation, an activation function, and a linear transformation. In this paper, we use $\text{ReLU}(x) = \max\{x, 0\}$ for all activation functions.

B. Reconstruction property of the proposed model

We prove that the proposed model has the *ideal learnable parameters* for periodic time series data. Consider D -dimensional time-series data $\{x_t\}$ with frequencies $f_1, \dots, f_M$. In other words, we assume that there exists a matrix $\widehat{W} \in \mathbb{R}^{D \times 2M}$ such that $\{x_t\}$ can be represented by

$$x_t = \widehat{W}(FE(t)). \quad (3)$$

Under the above assumption, the following proposition holds.

Proposition III.1. Let f denote the proposed model. Set the hyperparameters of f (i.e., the number of layers and dimensions of the hidden layers) arbitrarily. Then, there exist learnable parameters (i.e., weights and biases) such that $f(x_t) = x_t$, that is, f has the *ideal learnable parameters*.

Proof: By setting all the learnable parameters to 0 except for those in the output AFE layer, we have $f(x_t) = AFE(0, FE(t))$. We set the learnable parameters of the output AFE layer as follows. Here, we denote the L^∞ norm of a

TABLE I: Parameters of generating synthetic data

parameter	value
f	$[1/60, 1/360, 1/1440, 1/30, 1/720]$
A	$\begin{bmatrix} 1 & 3 & 2 & 0.5 & 0.3 \\ 2 & 1 & 3 & 0.5 & 0.3 \\ 2 & 1 & 3 & 0.5 & 0.3 \end{bmatrix}$
P	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 8 & 45 & 180 & 120 & 120 \\ 15 & 90 & 360 & 120 & 200 \end{bmatrix}$

TABLE II: Prepared dataset information

test dataset	injection interval
spike dataset	$[420, 425], [600, 605], [840, 845], [1080, 1085]$
level shift dataset	$[420, 480], [600, 660], [840, 900], [1080, 1140]$
ramp down dataset	$[420, 480], [600, 660], [840, 900], [1080, 1140]$
long down dataset	$[420, 480], [600, 660], [840, 900], [1080, 1140]$

vector by $\|\cdot\|_\infty$. Since $\sup_{t \in \mathbb{R}} \|\widehat{W}(FE(t))\|_\infty < \infty$, we can choose $b \in \mathbb{R}^D$ such that each component of b is larger than $\sup_{t \in \mathbb{R}} \|\widehat{W}(FE(t))\|_\infty$. Then, we have

$$\text{ReLU}(\widehat{W}(FE(t)) + b) = \widehat{W}(FE(t)) + b. \quad (4)$$

By setting $W_1 = 0, W_2 = I, W_3 = \widehat{W}, b_1 = b, b_2 = -b$ in (2), we have $AFE(0, FE(t)) = x_t$. Therefore, we obtain $f(x_t) = x_t$, which completes the proof of this proposition. ■

C. Characteristics of the proposed model

We describe two other characteristics of the proposed model besides the reconstruction property.

1) Flexibility: Even when time series data do not have a strict periodicity, the proposed model can perform reconstruction and anomaly detection by using its encoder-decoder structure. This represents a significant advantage of the proposed model over models that perform anomaly detection simply by Fourier transforming time series data, as those models cannot capture correlations within the data.

2) Computational cost: While the proposed model differs from existing encoder-decoder models only in the decoder component, the decoder's computational complexity remains $O(LH^2)$ for both training and inference when $M = O(H)$, where L denotes the number of decoder layers and H denotes the average hidden layer dimension. Moreover, the computational overhead of the FFT is negligible relative to model training complexity because the FFT is computed only once before training the model.

IV. EXPERIMENTS

We validated the effectiveness of the proposed model for anomaly detection by comparing it with existing encoder-decoder models using synthetic time series datasets in which the data have the *holes* and time-shifted datasets based on real-world data.

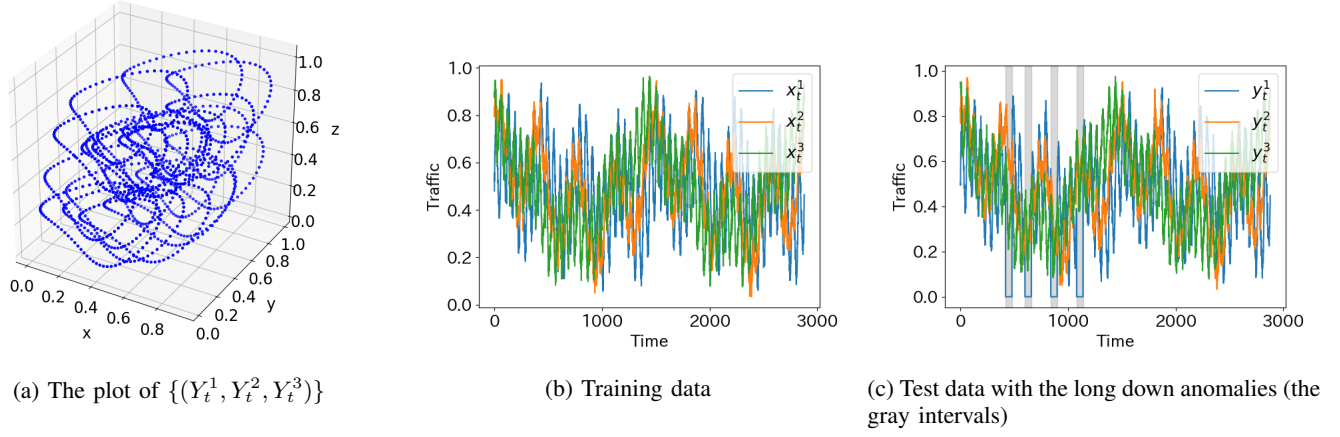


Fig. 3: Time series data used in Section IV-A

TABLE III: Reconstruction error of the training data in Section IV-A

model	error
AE	0.306 ± 0.008
AE-AFE (ours)	0.181 ± 0.003
1DCNN-AE	0.191 ± 0.009
1DCNN-FC	0.197 ± 0.005
1DCNN-AFE (ours)	0.181 ± 0.003
LSTM-AE	0.401 ± 0.053
LSTM-FC	0.495 ± 0.006
LSTM-AFE (ours)	0.169 ± 0.001
Transformer-AE	0.201 ± 0.006
Transformer-FC	0.072 ± 0.002
Transformer-AFE (ours)	0.164 ± 0.003

A. Synthetic datasets

In this experiment, we aim to show that the proposed model is effective for periodic time series data with the *holes*.

1) Datasets: By using the parameters as in Table I and mutually independent Gaussian random variables $\varepsilon_t^k \sim N(0, 0.03)$ for $k = 1, 2, 3$, the periodic time series data $\{(x_t^1, x_t^2, x_t^3)\}$ were created by the following equations:

$$X_t^k = \sum_{i=1}^5 A_{ki} \sin 2\pi f_i(t + P_{ki}) \quad (5)$$

$$(Y_t^1, Y_t^2, Y_t^3) = \text{Norm}(X_t^1, X_t^2, X_t^3) \quad (6)$$

$$(x_t^1, x_t^2, x_t^3) = \text{Norm}((Y_t^1, Y_t^2, Y_t^3) + (\varepsilon_t^1, \varepsilon_t^2, \varepsilon_t^3)), \quad (7)$$

where Norm denotes the normalizing operator to $[0, 1]^3$.

We can observe that $\{(x_t^1, x_t^2, x_t^3)\}$ has many *holes*, as shown in the plot of $\{(Y_t^1, Y_t^2, Y_t^3)\}$ in $\mathbb{R}^3$ presented in Figure 3 (a). Since the direct plot of $\{(x_t^1, x_t^2, x_t^3)\}$ in $\mathbb{R}^3$ is obscured by the noise ε_t^k , we instead plot $\{(Y_t^1, Y_t^2, Y_t^3)\}$ in $\mathbb{R}^3$, showing that $\{(Y_t^1, Y_t^2, Y_t^3)\}$ has many *holes*. Given that (x_t^1, x_t^2, x_t^3) and (Y_t^1, Y_t^2, Y_t^3) only differ by noise and normalization, we can conclude that $\{(x_t^1, x_t^2, x_t^3)\}$ also have many *holes*.

The training data were set to $\{(x_t^1, x_t^2, x_t^3)\}_{t=0}^{2879}$, which is shown in Figure 3 (b). Then, we set $L = 16$ and prepared the training dataset $\{(x_{t:t+L}^1, x_{t:t+L}^2, x_{t:t+L}^3)\}_{t=0}^{2864}$, where $x_{t:t+L}^k$ denotes $x_{t:t+L}^k = (x_t^k, \dots, x_{t+L-1}^k) \in \mathbb{R}^L$ for $k = 1, 2, 3$. Here, L denotes the window size, which is the number of consecutive data points input simultaneously as a time series.

The test data were created by injecting the following anomalies into $\{(y_t^1, y_t^2, y_t^3)\}_{t=0}^{2879}$, where $y_t^k = x_{t+2880}^k$ for $k = 1, 2, 3$. It is known from previous works [16], [17] that the main anomaly patterns in communication networks consist of the following four types:

- Spike: $\text{Spike}(x_t) = 0$
- Level shift: $\text{LS}_r(x_{t:t+K}) = rx_{t:t+K}$ ($0 < r < 1$)
- Ramp down: $\text{RD}_r(x_{t:t+K}) = (r^k x_{t+k})_{k=0}^{K-1}$ ($0 < r < 1$)
- Long down: $\text{LD}(x_{t:t+K}) = 0$

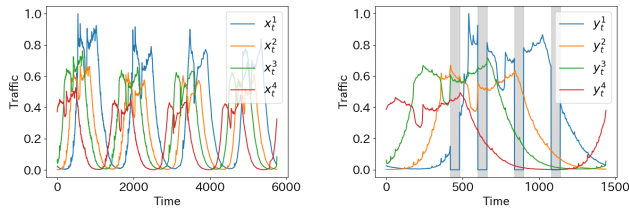
Based on the above anomaly types, we created four test data: spike data, level shift data ($r = 0.5$), ramp down data ($r = 0.95$), and long down data, by injecting each anomaly into y_t^1 at the time intervals listed in Table II. For example, the long down data was created by inserting the long down anomaly for $t \in [420, 480] \cup [600, 660] \cup [840, 900] \cup [1080, 1140]$, which is shown in Figure 3 (c). Then, four test datasets were created with a window size $L = 16$: spike dataset, level shift dataset, ramp down dataset, and long down dataset.

2) Models and experimental settings: As existing models, we prepared AE, x-AE, and x-FC, where x stands for 1DCNN, LSTM, or Transformer. x-AE is a model explained in Section II-A. x-FC is a model that has an encoder with x structure, while the decoder is a standard FC layer. We selected these models i) to demonstrate that the proposed model is effective for major types of encoder layers, and ii) to demonstrate that the AFE layer and $FE(t)$ are necessary for encoder-decoder models by comparing existing models with the proposed model. Since this paper focuses on improving encoder-decoder models, we did not implement other types of state-of-the-art anomaly detection models.

As the proposed models, we prepared AE-AFE and x-AFE. AE-AFE has an encoder with a standard FC layer and a

TABLE IV: AUC of anomaly detection results in Section IV-A

model	spike	level shift	ramp down	long down	average
AE	0.995 ± 0.014	0.830 ± 0.058	0.923 ± 0.040	0.998 ± 0.005	0.937
AE-AFE (ours)	1.000	1.000	0.972 ± 0.026	1.000	0.993
1DCNN-AE	1.000	0.977 ± 0.149	0.951 ± 0.016	1.000	0.982
1DCNN-FC	1.000	0.976 ± 0.125	0.950 ± 0.031	1.000	0.982
1DCNN-AFE (ours)	1.000	1.000	0.974 ± 0.034	1.000	0.994
LSTM-AE	0.980 ± 0.169	0.852 ± 0.093	0.890 ± 0.155	0.991 ± 0.104	0.928
LSTM-FC	0.973 ± 0.013	0.804 ± 0.024	0.841	0.990 ± 0.010	0.902
LSTM-AFE (ours)	1.000	1.000	0.975 ± 0.030	1.000	0.994
Transformer-AE	1.000	0.915 ± 0.106	0.932 ± 0.028	0.997 ± 0.077	0.961
Transformer-FC	1.000	0.916 ± 0.143	0.885 ± 0.020	1.000	0.950
Transformer-AFE (ours)	1.000	1.000	0.973 ± 0.043	1.000	0.993



(a) Training data

(b) Test data with the long down anomalies (the gray intervals)

Fig. 4: Time series data used in Section IV-B

decoder with the AFE layer. x-AFE has an encoder with x structure and a decoder with the AFE layer. To construct AE-AFE and x-AFE, we need to set $FE(t)$ in the AFE layer, which is defined by (1). We set $M = 3$ and define f_1 , f_2 , and f_3 as $\mathbf{f}_1$, $\mathbf{f}_2$, and $\mathbf{f}_3$.

The following are the experimental settings. In each model, the number of hidden layers and their dimensions were set so that the total number of learnable parameters was approximately 250,000. The number of epochs was 30 and the mini-batch size was 128. The optimization method was Adam [18] with a learning rate of 0.001. The loss function was the mean squared error function.

We conducted experiments 5 times with the above settings and calculated the average and standard deviation of the area under the curve (AUC).

3) Results: Table III shows the reconstruction error of the training data. We can observe that the proposed models achieved lower reconstruction error compared to the existing models, except when x is Transformer. This is because the proposed models have the *ideal learnable parameters*, although the existing models do not have the *ideal learnable parameters* due to the *holes* of the data. Therefore, the proposed models can obtain better latent representations of the data than the existing models.

As a result, the proposed models outperformed the existing models in terms of AUC scores, as shown in Table IV. In particular, LSTM-AFE outperformed the existing models LSTM-AE and LSTM-FC by 6.6% in terms of AUC scores.

TABLE V: Reconstruction error of the training data in Section IV-B

model	error
AE	0.093 ± 0.014
AE-AFE (ours)	0.087 ± 0.011
1DCNN-AE	0.064 ± 0.008
1DCNN-FC	0.077 ± 0.009
1DCNN-AFE (ours)	0.084 ± 0.006
LSTM-AE	0.106 ± 0.015
LSTM-FC	0.114 ± 0.015
LSTM-AFE (ours)	0.113 ± 0.011
Transformer-AE	0.119 ± 0.007
Transformer-FC	0.054 ± 0.006
Transformer-AFE (ours)	0.064 ± 0.007

B. Time-shifted datasets based on real-world data

In this experiment, we aim to demonstrate that the proposed model is effective for periodic time-shifted data.

1) Datasets: First, we prepared four-dimensional real traffic data collected from a communication network provided by the NTT group. The data were sampled at one-minute intervals and exhibit a one-day periodicity. Since there were some missing values in the data, we interpolated the data using linear interpolation. Next, the data were normalized to $[0, 1]^4$. Finally, we shifted each real traffic data, which we denote $\{\mathbf{x}_t\}$ or $\{(x_t^1, x_t^2, x_t^3, x_t^4)\}$.

The training data consisted of five consecutive weekdays from the data $\{\mathbf{x}_t\}$, which is shown in Figure 4 (a). Then, we created the training dataset with a window size $L = 16$.

In addition, we created four test data ($y_t^1, y_t^2, y_t^3, y_t^4$) by injecting anomalies into one weekday of the data $\{\mathbf{x}_t\}$. The injected anomalies and their intervals are listed in Table II, which are the same as those in Section IV-A. For example, Figure 4 (b) illustrates the long down test data. Then, we created four corresponding test datasets with a window size $L = 16$.

2) Models and experimental settings: We used the same models as in Section IV-A. To set $FE(t)$ in (1), we set $M = 6$ and determined the frequencies of the training data using the Fourier transform. Then, we used the top 6 largest frequencies in the data for $FE(t)$.

TABLE VI: AUC of of anomaly detection results in Section IV-B

model	spike	level shift	ramp down	long down	average
AE	0.995 ± 0.055	0.939 ± 0.070	0.968 ± 0.051	0.995 ± 0.007	0.974
AE-AFE (ours)	0.996 ± 0.026	0.961 ± 0.122	0.974 ± 0.032	0.996 ± 0.007	0.982
IDCNN-AE	0.998 ± 0.004	0.954 ± 0.069	0.975 ± 0.083	0.996 ± 0.018	0.981
IDCNN-FC	0.995 ± 0.006	0.952 ± 0.042	0.968 ± 0.060	0.995 ± 0.007	0.978
IDCNN-AFE (ours)	0.995 ± 0.010	0.958 ± 0.086	0.975 ± 0.013	0.995 ± 0.005	0.981
LSTM-AE	0.997 ± 0.005	0.952 ± 0.111	0.972 ± 0.067	0.996 ± 0.007	0.979
LSTM-FC	0.997 ± 0.011	0.940 ± 0.205	0.969 ± 0.093	0.994 ± 0.011	0.975
LSTM-AFE (ours)	0.997 ± 0.010	0.971 ± 0.076	0.981 ± 0.106	0.998 ± 0.002	0.987
Transformer-AE	0.995 ± 0.010	0.831 ± 0.125	0.938 ± 0.181	0.964 ± 0.163	0.932
Transformer-FC	1.000 ± 0.015	0.976 ± 0.208	0.980 ± 0.198	1.000	0.989
Transformer-AFE (ours)	1.000 ± 0.052	0.987 ± 0.117	0.985 ± 0.044	1.000	0.993

In each model, the number of hidden layers and their dimensions were set so that the total number of learnable parameters was approximately 450,000. The other settings were the same as in Section IV-A.

3) Results: Table V shows the reconstruction error of the training data. We can see that AE-AFE achieved lower reconstruction error than AE, while the other proposed models resulted in larger reconstruction errors than their corresponding existing models. This is because, although the proposed models have the *ideal learnable parameters*, there is no guarantee that the proposed models can obtain the *ideal learnable parameters* through optimization.

Table VI illustrates the AUC of each model on the four test datasets, indicating that the proposed models achieved better performance on average in terms of AUC than the existing models, despite having larger reconstruction errors. This result shows that the proposed models can be effective in anomaly detection for real time-shifted data.

The high performance of the proposed models in anomaly detection is explained as follows. The proposed models can perform anomaly detection by not only capturing the correlations within data, but also by using the frequencies of data, whereas the existing models only capture the correlations within data. Therefore, in anomaly detection, the proposed model can become more sensitive to anomalous data that deviate from periodicity than existing models.

V. CONCLUSION

In this paper, we proposed the encoder-decoder model with the *ideal learnable parameters* for periodic time series data. We pointed out that existing encoder-decoder models have a limitation in reconstructing input data depending on the geometric features of the data, such as *holes*. To enable the proposed model to have the *ideal learnable parameters*, we introduced the AFE layer, which incorporates the frequencies and correlations within periodic time series data. In anomaly detection experiments with two periodic time series datasets, we showed that the proposed model achieved better AUC scores than existing encoder-decoder models.

For future work, we will further improve the proposed model to handle not only periodic time series data, but also stationary time series data.

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